

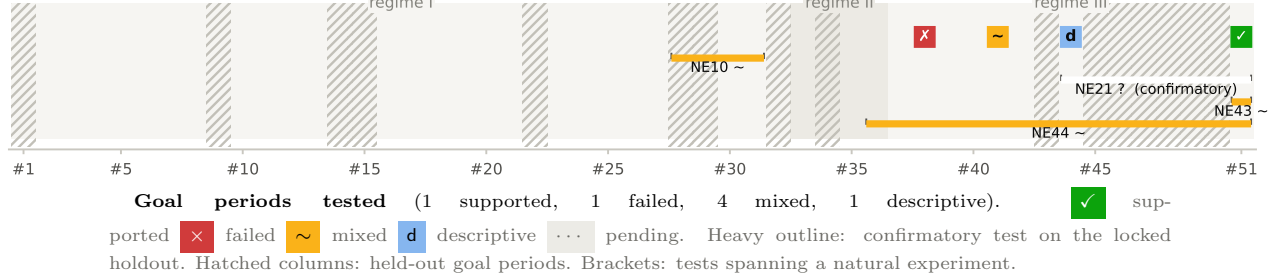
H04: External forcing reshapes the response kernel, reversibly

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 1.5/5 Usefulness 2.5/5

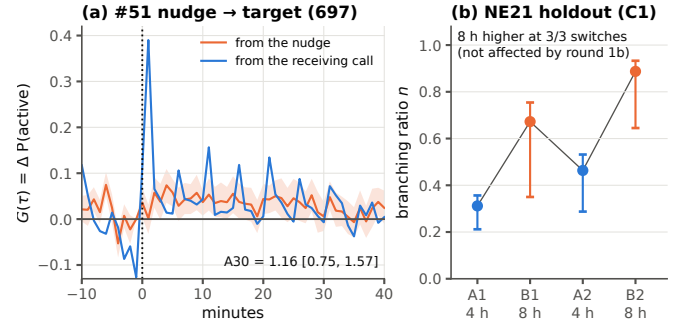
Bottom line. A nudge acts only on its target at the target's next model call; the dead time was an artifact, and no hours or nudger switch lowers self-excitation.

level: hypothesis · pipeline checks 98% · updated 2026-10-04



WHAT HAPPENED

Round 1b is exploratory: non-holdout days, the corrected activity table, leading-@ targets, no future-kick isolation and a day fixed effect. In #51 a nudge adds 1.16 [0.75, 1.57] active minutes to its named target over 30 min. First nudges add 1.38, repeats 0.90; regime-III 4 h days give 0.93. By-standers gain ≈ 0 . The response starts at the target's next model call: the median read-out is 108 s, and the onset is 1 min after read-out. Dropped events plus the isolation rule made round 1's 4-min dead time. Human messages no longer move activity. The FDT fails in shape, and kicked activations persist ($X \approx 1.9$). Mean-field Glauber underpredicts the decay times. With the nudger off (NE43), n stays the same and traps do not lengthen. **The locked holdout NE21** switches the hours 4 $\rightarrow$ 8 $\rightarrow$ 4 $\rightarrow$ 8 h. **C1 fails in reverse:** n is 0.31, 0.67, 0.46, 0.89. C1 does not use the corrected table, so the result stands.



Observables. (a) A nudged agent's activity against matched controls (corrected design, 95% day-block band). The curves align on the nudge and on the target's receiving call. On the receiving call the response is immediate. The delay from the nudge is the wait until the target's next call. (b) The branching ratio per hours segment on the holdout. Every switch goes the other way, at the edge of week-to-week noise.

PHYSICAL MODEL

Models 02 (linear response, fluctuation-dissipation) and 09 (Hawkes). $n_i(m) \in \{0, 1\}$ says whether agent i is active in minute m . A kick (a nudge or a human message) at m_0 is a field pulse. G is the matched response, and r is the read-out delay to the target's next model call. C is the activity autocorrelation, χ the integrated G and X the FD ratio. In the Hawkes fit, $\mu_d b(t)$ is the daily baseline and ϕ the talk-triggered kernel. η_c counts messages triggered per class- c kick, and n is the branching ratio. In mean field, K is the loop gain and τ_0 the agent time.

$$G(\tau) = \mathbb{E}[n_i(m_0 + \tau) \mid \text{kick}] - \mathbb{E}[n_i(m_0 + \tau) \mid \text{control}] - \delta_d(\tau)$$

$$G(\tau) \approx \sum_r p(r) \Delta R(\tau - r), \quad \text{FDT: } \chi(\tau) = X [C(0) - C(\tau)]$$

$$\lambda(t) = \mu_d b(t) + \sum_k \phi(t - t_k) + \sum_c \eta_c \sum_j \gamma_c e^{-\gamma_c(t - t_j)}$$

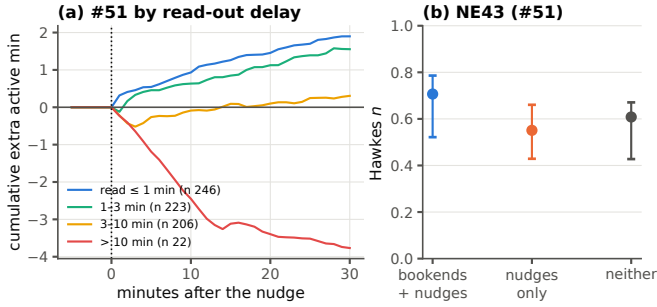
$$n = \int_0^\infty \phi(\tau) d\tau, \quad \tau = \tau_0 / (1 - K)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A nudge acts on the agent it names and on nobody else. It acts at that agent's next model call, so the target's pause schedule sets its lag. A nudge to an agent paused for 10 min does nothing until the agent wakes. Send a nudge to an awake agent: a first nudge buys about 1.4 active minutes, a repeat 0.9. Human messages change what agents say, not how much they work. The physics vocabulary (Green's functions, FDT, effective temperature) mostly shows that a field-on-activity model is wrong. The one reversal test fails in the opposite direction: longer days raise self-excitation. With the nudger off, self-excitation stays the same.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
P1 nudge: A60 > 0, peak ~15 min → #51 peak 4, peak ≤5 min A60 1.65 [0.79, 2.52]	yes (#51, 4 h)	
Nudge A30 (isolated → 1.54 → 1.16 (#51), 0.93 (4 h); all nudges) isolation inflates ×1.3–1.9	corrected	
First vs repeat nudge – → 1.38 vs 0.90 (#51)	sub-additive	
P2 bystanders ≤0.2 of yes → 0.03 (#51) target	yes	
P3 human messages mentioned 5.6 → 0.6 (n.s.); room move activity ≈ 0	no	
P6 FDT violated, violated, X 1.54 → 1.88 nudge X < 1	violated; dir. no	
MF-P1/P3 (K subcrit- K III 0.26 → 0.32 ical, self-consistent)	yes	
NE43 nudger off: n Δn +0.06; inactive runs +0.5% same, traps longer	mixed	
NE44 read-out moves 43 s vs 108 s; onset lag equal from with pause default read-out	mixed	
C1 (holdout) n lower 0/3 switches; not affected by 1b on 8 h days	falsified	



Second figure. (a) The cumulative response of #51 nudge targets, by how long the nudge waits to be read. Read within 3 min, the response starts at once. Read after 10 min (long pauses), the target stays below its controls. (b) NE43: Hawkes n with the daily bookends and nudges, with nudges only, and with neither.

GOAL PERIODS IN DETAIL

NE21 ? 1b: not re-run (locked holdout; re-running is Vivian'... (holdout))

G38 × failed
G41 ~ mixed
G44 d descriptive
G51 ✓ supported
NE10 ~ mixed
NE43 ~ mixed
NE44 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The #51 pre-window placebo is +0.25 [0.05, 0.46]. Some selection by the nudger's text trigger remains, so the level can be up to a quarter too high. The matching strata do not include long pauses. Regime-I nudges (NE10) fail the placebo. Hawkes n can absorb within-day non-stationarity. The holdout's C2, C4 and MF-C used the buggy activity table and the isolation rule. Vivian decides whether to re-run them.

ROUND 2

Add the pause state to the matching strata. Fit a read-out plus transient kernel.

- **Essence.** The scaffold is the propagator: the response to any input is when the agent next looks, times whether it attends.
- **H04-R1** The response kernel equals the turn-interval survival function convolved with an uptake probability, with no free parameters (HH92, tested in H08).
- **H04-R2** The steering knob is the scheduler: shortening turn cadence raises susceptibility more than rewording or repeating a message (E1).
- **H04-R3** A salience law: uptake rises with mention, position and novelty in context and falls with backlog (E5).